

Quiz 6

1.) Which are isometries? A glide refl.

C id. map

homotheties, contractions change lengths.

2) direct isometries? $\Rightarrow \det A = 1$.

A rotations.

B composition of reflections. $(-1) \cdot (-1) = 1$.

3) which are glide-reflections?

$$a) \det \left(\frac{1}{5} \begin{pmatrix} 4 & -3 \\ 3 & 4 \end{pmatrix} \right) = \frac{1}{5^2} \begin{vmatrix} 4 & -3 \\ 3 & 4 \end{vmatrix} = \frac{1}{5^2} (16 + 9) = \frac{25}{25 \cdot 1} = 1. \quad \text{direct.}$$

$$b) \det \frac{1}{5^2} \begin{pmatrix} 4 & 3 \\ 3 & -4 \end{pmatrix} = \frac{1}{25} \cdot (-16 - 9) = -1 \Rightarrow \text{indir.} \quad \checkmark \quad \boxed{B}$$

$$c) \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} = 0 + 1 = 1$$

$$d) \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0 - 1 = -1 \quad \boxed{D}$$

Setup I: $\mathbb{R}^2$, $P(0,1)$, $l: x+2y-1=0$

$\text{Ref}_l^\perp$

$$\phi = \text{Rot}_{0,45^\circ} \circ \text{Ref}_l^\perp$$

$\text{Rot}_{0,45^\circ}$

$$P' = P - 2 \cdot \frac{ax_0 + by_0 + c}{a^2 + b^2} (a, b)$$

orthogonal ref. formula.

$$a=1, b=2, c=-1.$$

4) $\text{Ref}_l^\perp(P)$; $P(0,1)$

$$ax_0 + by_0 + c = 1 \cdot 0 + 2 \cdot 1 - 1 = 1$$

$$a^2 + b^2 = 1^2 + 2^2 = 5$$

$$P' = P - 2 \cdot \frac{1}{5} \cdot (1, 2) = (0, 1) - \frac{2}{5} (1, 2) = \left(-\frac{2}{5}, \frac{1}{5} \right) \quad \boxed{A}$$

5) $\phi(O) = \text{Rot}_{0,45^\circ} \circ (\text{Ref}_l^\perp(O))$, $O(0,0) \Rightarrow x_0=0, y_0=0$

$$\text{Ref}_l^\perp(O): ax_0 + by_0 + c = a \cdot 0 + b \cdot 0 + (-1) = -1.$$

$$O' = (0, 0) - 2 \cdot \frac{(-1)}{5} (1, 2) = \frac{2}{5} (1, 2) = \left(\frac{2}{5}, \frac{4}{5} \right)$$

$$\text{Rot}_{0,45^\circ}: R = \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$R(O') = R \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \end{pmatrix} = \frac{\sqrt{2}}{2} \cdot \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} \frac{2}{5} - \frac{4}{5} \\ \frac{2}{5} + \frac{4}{5} \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} -\frac{2}{5} \\ \frac{6}{5} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{5} \\ \frac{3\sqrt{2}}{5} \end{pmatrix} \doteq \phi(O').$$

6. $\phi^{-1} = ?$

$$\phi(x) = Ax + b.$$

$$\text{Ref}_l^\perp(x) = Sx + d.$$

$$l: x + 2y - 1 = 0, \quad m = (1, 2) \Rightarrow \boxed{S = I - 2 \cdot \frac{mm^T}{m^T m}}$$

$$\begin{matrix} 2 \times 1 & \times & 1 \times 2 \\ 1 \times 2 & \times & 2 \times 1 \end{matrix}$$

$$mm^T = \begin{pmatrix} 1 \\ 2 \end{pmatrix} (1 \ 2) = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad m^T m = (1 \ 2) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1 + 4 = 5.$$

$$S = I - \frac{2}{5} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} \frac{2}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{8}{5} \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & -\frac{3}{5} \end{pmatrix}$$

$$\text{Translation part: } \boxed{d = -2 \cdot \frac{c}{m^T m} m}$$

$$d = -2 \cdot \frac{-1}{5} (1, 2) = \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \end{pmatrix}$$

$$x \mapsto Sx + d$$

$$\phi'(x) = R(Sx + d) = \underbrace{RS}_A x + \underbrace{Rd}_B = \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & -\frac{3}{5} \end{pmatrix} x + \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \end{pmatrix}$$

$$R \stackrel{5)}{=} \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$2 \times 2 \quad 2 \times 2$$

$$\begin{aligned} \phi(x) &= \frac{\sqrt{2}}{10} \begin{pmatrix} \frac{3}{5} + \frac{4}{5} & -\frac{4}{5} + \frac{3}{5} \\ -\frac{4}{5} - \frac{4}{5} & -\frac{4}{5} - \frac{3}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \frac{\sqrt{2}}{10} \begin{pmatrix} \frac{2}{5} - \frac{4}{5} \\ \frac{2}{5} + \frac{4}{5} \end{pmatrix} \\ &= \frac{\sqrt{2}}{10} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \frac{\sqrt{2}}{10} \begin{pmatrix} -\frac{2}{5} \\ \frac{6}{5} \end{pmatrix} \dots \end{aligned}$$

$$\phi(x) = Ax + b$$

$$\phi'(x) = A^{-1}x - A^{-1}b$$

$$A^{-1} = A^T \quad (A \text{ orthogonal})$$

$$\mathbb{H}^2, P(2,1), C(-1,1)$$

Rot_{C,θ} around C by θ, cos θ = $\frac{3}{5}$, counterclockwise ⇒ sin θ = $\frac{4}{5}$

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$$

$$\Phi = \text{Rot}_{O, \tilde{\theta}} \circ \text{Rot}_{C, \theta}$$

$$7) \text{Rot}_{C, \theta} \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \dots$$

Rotation around C ⇒ Rot_{C,θ}(x) = R_θ(x - C) + C = R_θx + (C - R_θC)

$$\Rightarrow \text{Rot}_{C, \theta} \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{5} \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

$$C - R_\theta C = \begin{pmatrix} -1 \\ 1 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} -7 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 + \frac{7}{5} \\ 1 + \frac{1}{5} \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ \frac{6}{5} \end{pmatrix}$$

$$8) \tilde{\theta} = -\theta, \Phi(P) = ?$$

$$\Phi = \text{Rot}_{O, -\theta} \circ \text{Rot}_{C, \theta}$$

$$R_{-\theta} = R_\theta^{-1} = R_\theta^T = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}$$

$$(7) \Rightarrow \boxed{\text{Rot}_{C, \theta}(x) = R_\theta x + b}$$

$$\Phi(x) = R_{-\theta}(R_\theta x + b) = x + R_{-\theta}b \Rightarrow \text{translation.}$$

$$R_{-\theta}b = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix} \cdot \frac{1}{5} \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \frac{1}{25} \begin{pmatrix} 6 + 24 \\ -8 + 18 \end{pmatrix} = \frac{1}{25} \begin{pmatrix} 30 \\ 10 \end{pmatrix} = \begin{pmatrix} \frac{6}{5} \\ \frac{2}{5} \end{pmatrix}$$

$$\Phi(x) = x + \begin{pmatrix} \frac{6}{5} \\ \frac{2}{5} \end{pmatrix}$$

$$P(2,1)$$

$$\Phi(2,1) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{6}{5} \\ \frac{2}{5} \end{pmatrix} = \begin{pmatrix} \frac{16}{5} \\ \frac{7}{5} \end{pmatrix}$$

$$9) \tilde{\theta} = -\theta, \Phi \text{ translation } \boxed{A} \quad \boxed{B}$$

10. $\mathbb{R}^2$, $c_1(1,0)$, $c_2(-1,0)$

Rot c_1, θ around c_1 , counterclockwise, $\cos \theta = \frac{3}{5}$, $\sin \theta = \frac{4}{5}$

$$\Phi = \text{Rot}_{c_2, \tilde{\theta}} \circ \text{Rot}_{c_1, \theta}$$

10) O origin, $\tilde{\theta} = 90^\circ$, $\Phi(O) = \dots$

$$R_\theta = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}$$

$$R_{90} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\Phi(O) = \text{Rot}_{c_2, 90^\circ} \circ (\text{Rot}_{c_1, \theta}(O))$$

$$\hookrightarrow \boxed{R_\theta (x - c_1) + c_1} = \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} x-1 \\ y-0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

$$\xrightarrow{x=y=0} \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

$$= \frac{1}{5} \begin{pmatrix} -3 \\ -4 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} + 1 \\ -\frac{4}{5} \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ -\frac{4}{5} \end{pmatrix} = Q$$

rotate this by 90°

$$\boxed{R_{90}(Q - c_2) + c_2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{2}{5} + 1 \\ -\frac{4}{5} - 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{6}{5} \\ \frac{7}{5} \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{1}{5} \\ \frac{7}{5} \end{pmatrix} = \Phi(O).$$

11) $\tilde{\theta} = 90^\circ$, Φ rotation with center $\dots$

$$\Phi = \text{Rot}_{c_2, 90^\circ} \circ \text{Rot}_{c_1, \theta} \Rightarrow \text{lin. part: } R_{90} R_\theta = A \Rightarrow \text{angle } 90^\circ + \theta. \Rightarrow$$

$$\Rightarrow \text{rotation.} \Rightarrow \Phi(M) = M \text{ (center } M)$$

$$\Phi(x) = Ax + b$$

$$x = Ax + b$$

center satisfies

$$\Rightarrow (I - A)x = b$$

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -4 & -3 \\ 3 & -4 \end{pmatrix}$$

$$\Phi(x) = R_{90}(R_\theta x + b_1) + b_2 = Ax + \underbrace{R_{90}b_1 + b_2}_b$$

lin. part transl. part.

$$R_\theta(X - c_1) + c_1 = R_\theta X + \underbrace{(c_1 - R_\theta c_1)}_{b_1}$$

$$R_{g_0^\circ}(X - c_2) + c_2 = R_{g_0^\circ} X + \underbrace{(c_2 - R_{g_0^\circ} c_2)}_{b_2}$$

$$b_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 - \frac{3}{5} \\ 0 - \frac{4}{5} \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ -\frac{4}{5} \end{pmatrix}$$

$$b_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$b = R_{g_0^\circ} b_1 + b_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{2}{5} \\ -\frac{4}{5} \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ \frac{2}{5} \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} \\ \frac{7}{5} \end{pmatrix}$$

$$(I - A)X = b$$

$$\Rightarrow \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} -4 & -3 \\ 3 & -4 \end{pmatrix} \right) X = \begin{pmatrix} -\frac{1}{5} \\ \frac{7}{5} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 + \frac{4}{5} & \frac{3}{5} \\ -\frac{3}{5} & 1 + \frac{4}{5} \end{pmatrix} X = \dots$$

$$\Rightarrow \begin{pmatrix} \frac{9}{5} & \frac{3}{5} \\ -\frac{3}{5} & \frac{9}{5} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} \\ \frac{7}{5} \end{pmatrix} \Rightarrow \begin{cases} \frac{9}{5}x + \frac{3}{5}y = -\frac{1}{5} \\ -\frac{3}{5}x + \frac{9}{5}y = \frac{7}{5} \end{cases} \Rightarrow \begin{cases} 9x + 3y = -1 \\ -3x + 9y = 7 \end{cases} \quad 2)$$

$$2) \begin{cases} 9x + 3y = -1 \\ -3x + 9y = 7 \end{cases} \Rightarrow \begin{cases} 9x + 3y = -1 \\ -9x + 27y = 21 \end{cases} \quad (+)$$

$$30y = 20$$

$$y = \frac{2}{3}, 9x = -1 - 2 = -3 \Rightarrow x = -\frac{1}{3}$$

$$\Rightarrow \begin{pmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$$

$$12) \tilde{\theta} = ? \text{ s.t. } \phi \text{ translation?} \rightarrow \begin{matrix} I_3 = A \\ b \neq 0 \end{matrix}$$

$$\phi \rightarrow \text{total angle: } \tilde{\theta} + \theta$$

$$\text{for translation } \Rightarrow \tilde{\theta} + \theta = 0 \Rightarrow \tilde{\theta} = -\theta \quad \boxed{A}$$

Quiz 6 - redone

1) isometries $\rightarrow$ A glide-ref.
C id. map.

2) dir. isom: A rotations
B 2 refl.

3) glide ref?

$$\det \dots = -1 \Rightarrow \text{B, D.}$$

I. $\mathbb{E}^2$, $P(0,1)$, $l: x+2y-1=0 \Rightarrow (1,2)=a$, $a_{n+1}=-1$.

$\text{Ref}_l^\perp$ orthog. refl.

$\text{Rot}_{0,45^\circ}$ counterclockwise rot. around origin.

$$\Phi = \text{Rot} \circ \text{Ref}$$

$$4) \left[\text{Ref}_l^\perp(P) = P - 2 \cdot \frac{a^T P + a_{n+1}}{|a|^2} \cdot a \right]$$

$$P(0,1)$$

$$\Rightarrow \text{Ref} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 2 \cdot \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{2}{5} \\ \frac{1}{5} \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \text{A}$$

$$a^T P + a_{n+1} = (1 \ 2) \begin{pmatrix} 0 \\ 1 \end{pmatrix} - 1 = 2 - 1 = 1.$$

$$|a|^2 = \sqrt{2^2 + 1^2} = 5.$$

$$5) \Phi(0) = ?$$

$$\Phi = \text{Rot}_{0,45^\circ} \left(\text{Ref}_l^\perp(0) \right) = \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} \cdot \begin{bmatrix} \frac{2}{5} \\ \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{2}{5} \\ \frac{1}{5} \end{bmatrix} =$$

$$\text{Ref}_l^\perp(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 2 \cdot \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{2}{5} \\ \frac{4}{5} \end{bmatrix} = \begin{bmatrix} \frac{2}{5\sqrt{2}} & -\frac{4}{5\sqrt{2}} \\ \frac{2}{5\sqrt{2}} & +\frac{4}{5\sqrt{2}} \end{bmatrix}$$

$$a^T 0 + a_{n+1} = (1 \ 2) \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 1 = -1$$

Rotation not centered at the origin:

1) translate to origin

2) rotate at origin

3) translate back

$$\boxed{\text{Rot}_{e,45^\circ}(x) = R_{45^\circ}(x-c) + c}$$

2x2 2x1

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{2}{5} \\ \frac{4}{5} \end{bmatrix} = \Phi(0).$$

$$6) \quad \phi(x) = Ax + b$$

$$\phi^{-1}(x) = A^{-1}x - A^{-1}b$$

$$\text{Rot}_{0,45^\circ} = Ax + b = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \end{bmatrix} \checkmark$$

$$\text{Ref}_L^\perp = Cx + D =$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 2 \cdot \frac{2}{5} \right) x + 2 \cdot \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$a = (1, 2) ; \quad a \otimes a = a \cdot a^T = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = 1 + 4 = 5 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

$$|a|^2 = (\sqrt{5})^2 = 5.$$

$$= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{2}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \right) x + \frac{2}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \left(\frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix} \right) x + \frac{1}{5} \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$= \frac{1}{5} \cdot \begin{bmatrix} 3 & -4 \\ -4 & -3 \end{bmatrix} x + \frac{1}{5} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \checkmark$$

$$\phi = \text{Rot} \circ \text{Ref} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} (Cx + D)$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \left(\frac{1}{5} \begin{bmatrix} 3 & -4 \\ -4 & -3 \end{bmatrix} x + \frac{1}{5} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right)$$

$$= \frac{1}{5\sqrt{2}} \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix} x + \frac{1}{5\sqrt{2}} \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$= \frac{1}{5\sqrt{2}} \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix} x - \frac{2\sqrt{2}}{5\sqrt{2}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 7 & -1 & : & 1 & 0 \\ -1 & -7 & : & 0 & 1 \end{bmatrix} \xrightarrow{7R_2 + R_1 \rightarrow R_1} \begin{bmatrix} 0 & -15 & : & 1 & 7 \\ -1 & -7 & : & 0 & 1 \end{bmatrix}$$

$$\phi = \text{Rot} \circ \text{Ref} = A(Cx + D) + b = \underbrace{AC}x + \underbrace{AD + b}_0 = ACx + Ad$$

$$\phi = \frac{\sqrt{2}}{10} \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix} + \frac{\sqrt{2}}{5} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = Hx + b$$

$$\phi^{-1} = H^{-1}x - H^{-1}b = \frac{\sqrt{2}}{10} \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix} x + \frac{2}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$H = AC = \text{orthog. matrix} \Rightarrow H^{-1} = H^T = (AC)^T = \frac{\sqrt{2}}{10} \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix}$$

$$\begin{aligned} -H^{-1}b &= -\frac{\sqrt{2}}{10} \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix} \cdot \frac{\sqrt{2}}{5} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = -\frac{2}{5 \cdot 10} \cdot \begin{bmatrix} 7 & -1 \\ -1 & -7 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \\ &= -\frac{1}{25} \begin{bmatrix} -10 \\ -20 \end{bmatrix} = +\frac{10}{25} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{2}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{aligned}$$

$$I. \mathbb{E}^2, P(2,1), C(-1,1)$$

$$\text{Rot}_{C,\theta} \text{ around } C, \cos \theta = \frac{3}{5} \Rightarrow \sin \theta = \frac{4}{5}$$

$$\text{Rot}_{O,\tilde{\theta}} \text{ around } O,$$

$$\phi = \text{Rot}_{O,\tilde{\theta}} \circ \text{Rot}_{C,\theta}$$

$$\text{Rot. around } C: \text{Rot}_{C,\theta}(x) = R_{\theta}(x-C) + C = R_{\theta}x + (C - R_{\theta}C)$$

$$R_{\theta} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}$$

$$7) \text{Rot}_{C,\theta} = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x+1 \\ y-1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \left(\begin{bmatrix} -1 \\ 1 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right)$$

$$= \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} x + \left(\begin{bmatrix} -1 \\ 1 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} -4 \\ -1 \end{bmatrix} \right)$$

$$= \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} x + \frac{1}{5} \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

$$8) \tilde{\theta} = -\theta, \phi(P) = ?$$

$$R_{O,\tilde{\theta}} = \begin{bmatrix} \cos \tilde{\theta} & -\sin \tilde{\theta} \\ \sin \tilde{\theta} & \cos \tilde{\theta} \end{bmatrix} x = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} x = \begin{bmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{bmatrix} x$$

$$\phi = \text{Rot}_{O,\tilde{\theta}}(\text{Rot}_{C,\theta}(P)) = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix} \left(\frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right) =$$

$$= \frac{1}{5} \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix} \left(\frac{1}{5} \left(\begin{bmatrix} 2 \\ 11 \end{bmatrix} + \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right) \right)$$

$$= \frac{1}{5} \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 12 + 4 \cdot 17 \\ -16 + 3 \cdot 17 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 80 \\ 35 \end{bmatrix} = \frac{5}{25} \begin{bmatrix} 16 \\ 7 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 16 \\ 7 \end{bmatrix}$$

$$\begin{array}{r} 2 \\ 17 \cdot 4 \\ \hline 68 + \\ \hline 12 \\ \hline 80 \end{array}$$

$$\begin{array}{r} 2 \\ 17 \cdot 3 \\ \hline 51 - \\ \hline 16 \\ \hline 35 \end{array}$$

$$\begin{array}{r} 80 \mid 5 \\ 5 \mid 16 \\ \hline 30 \end{array}$$

9) $\tilde{\theta} = -\theta$,

$\text{Rot} \circ \text{Rot}$

$(1) \cdot (1) = (1) \Rightarrow \text{dir. isom } \boxed{B} \quad 100\%$

$\dots \quad \tilde{I}_3 \neq A$
 $b \neq 0 \Rightarrow \text{Translation } \boxed{A}$

III. $\mathbb{E}^2, c_1(1,0), c_2(-1,0)$

$\text{Rot}_{c_1, \theta} \text{ around } c_1 \Rightarrow R_\theta(x - c_1) + c_1$

$\text{Rot}_{c_2, \tilde{\theta}} \text{ around } c_2 \Rightarrow R_{\tilde{\theta}}(x - c_2) + c_2$

$\phi = \text{Rot}_{c_2} \circ \text{Rot}_{c_1}$

$= R_{\tilde{\theta}}(R_\theta(x - c_1) + c_1) + c_2$

$= R_{\tilde{\theta}}R_\theta(x) - R_{\tilde{\theta}}R_\theta c_1 + \underbrace{R_{\tilde{\theta}}c_1 + c_2}_b$

$= \dots$

10) $\phi(0) = \dots \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{bmatrix} \cdot \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix} x - \dots$

$\tilde{\theta} = 90^\circ \Rightarrow \phi \text{ rotation} \Rightarrow R_{\tilde{\theta}} \cdot R_\theta = I_2$

$\Rightarrow \frac{1}{5} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

2) $\frac{1}{5} \begin{bmatrix} -4 & -3 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A = \dots$
 $\Rightarrow -\frac{4}{5}x - \frac{3}{5}y = 1$
 $\frac{3}{5}x - \frac{4}{5}y =$

$\boxed{\phi(H) = H}$ H center. $\Rightarrow (I - A)x = b$.

$\Rightarrow \begin{bmatrix} \frac{9}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{9}{5} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1/5 \\ 7/5 \end{bmatrix} \Rightarrow \begin{cases} 9x + 3y = -1 \\ -9x + 27y = 21 \end{cases}$

$H(-\frac{1}{3}, \frac{2}{3})$

$x = -\frac{1}{3}$
 $y = \frac{2}{3}$

12) $\tilde{\theta} = ?$ s.t. ϕ translation?

$A = I_3, b \neq 0$

$\phi \rightarrow \text{total angle: } \tilde{\theta} + \theta = 0 \Rightarrow \tilde{\theta} = -\theta \quad \boxed{A}$